

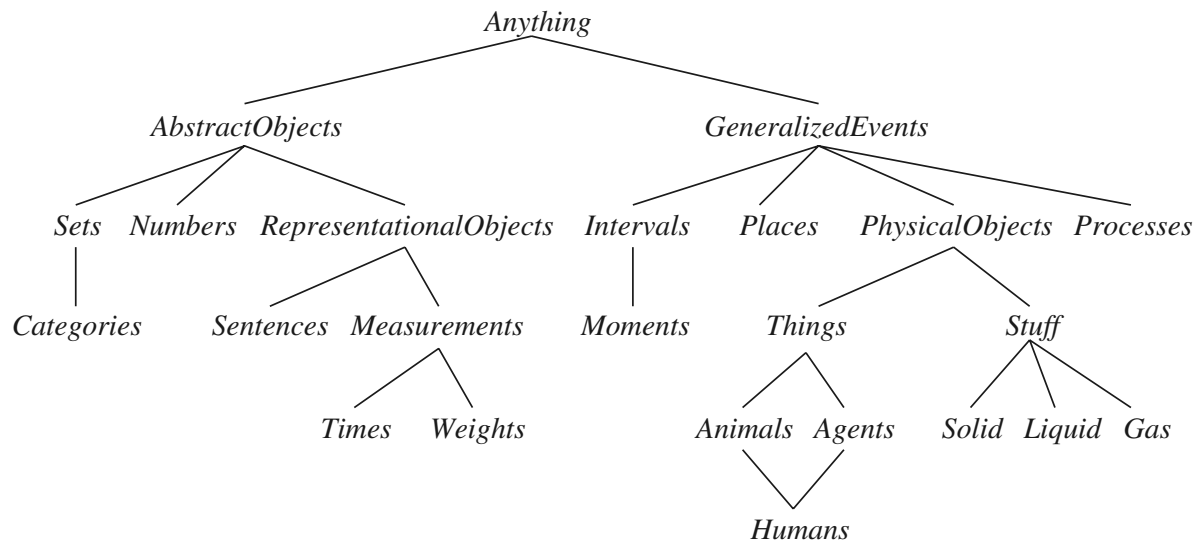
Artificial Intelligence

Knowledge Representation

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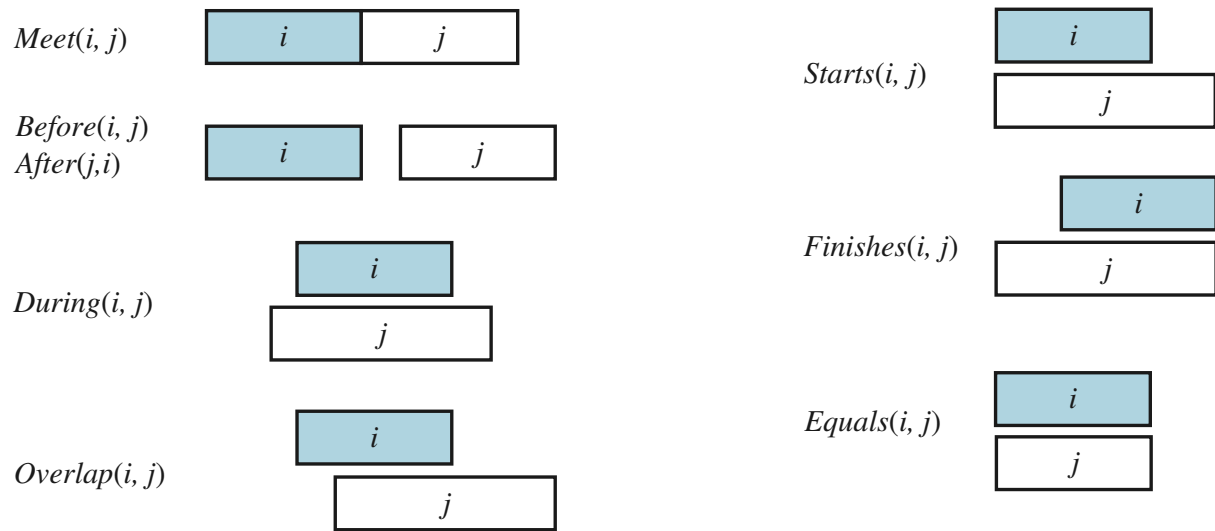
1 Ontological Engineering



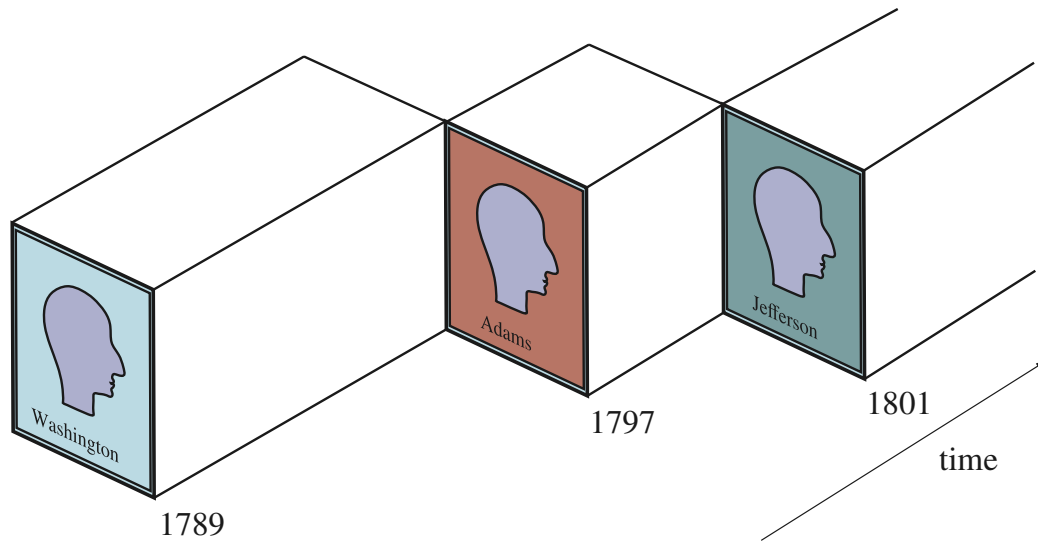
2 Categories and Objects

Foo

3 Events



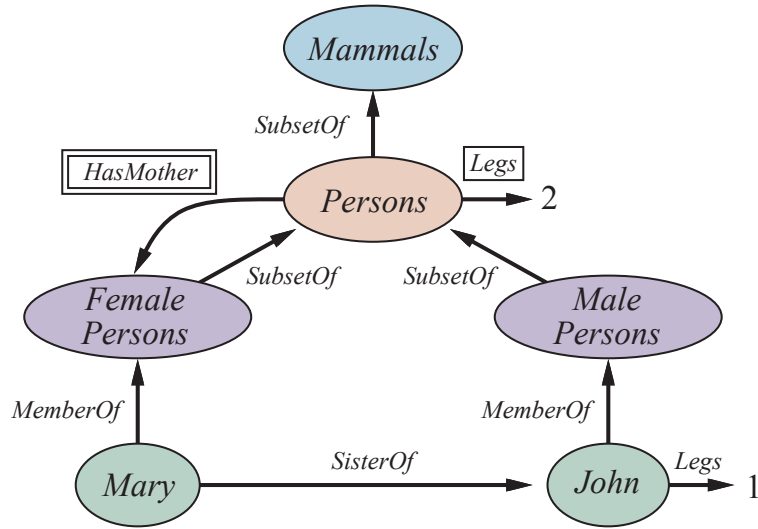
4 Fluents



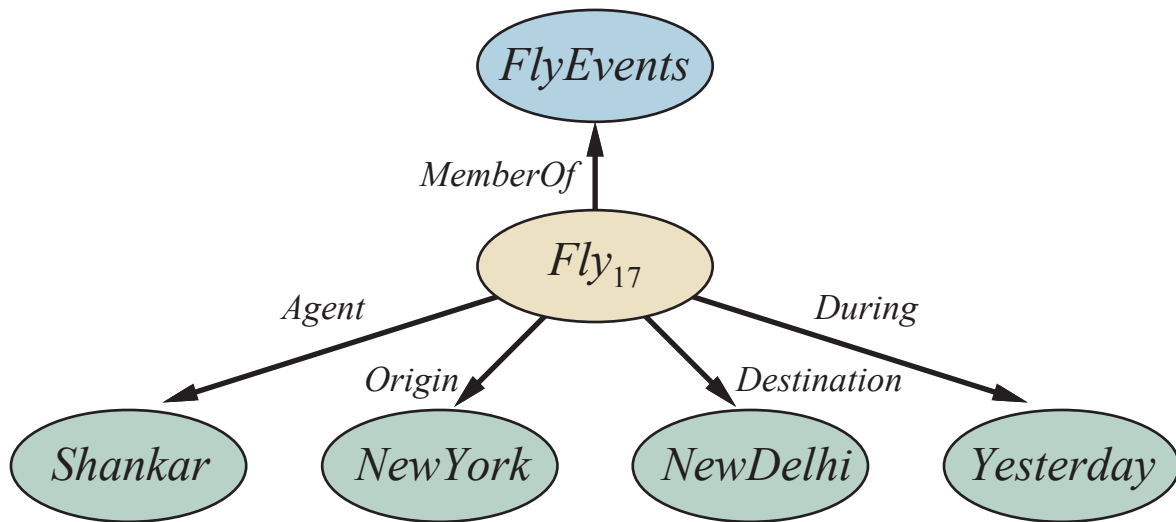
5 Mental Objects and Modal Logic

Foo

6 Reasoning Systems for Categories



7 Reasoning Systems for Categories



8 Description Logics

$Concept \rightarrow \mathbf{Thing} \mid ConceptName$
| $\mathbf{And}(Concept, \dots)$
| $\mathbf{All}(RoleName, Concept)$
| $\mathbf{AtLeast}(Integer, RoleName)$
| $\mathbf{AtMost}(Integer, RoleName)$
| $\mathbf{Fills}(RoleName, IndividualName, \dots)$
| $\mathbf{SameAs}(Path, Path)$
| $\mathbf{OneOf}(IndividualName, \dots)$
 $Path \rightarrow [RoleName, \dots]$
 $ConceptName \rightarrow Adult \mid Female \mid Male \mid \dots$
 $RoleName \rightarrow Spouse \mid Daughter \mid Son \mid \dots$

9 Reasoning with Default Information

Truth maintenance systems